
UCL COMP0078: Supervised Learning

Problem Sheet

First Order Optimization & ERM

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Overview of Problems

- Problem 1: On lecture slide 31 (Theorem GD rates), it is shown that, provided that a function is convex and smooth, gradient descent ‘works’ in the sense that $|F(w_K) - F(w_*)|$ tends to zero when the number of iterations K tends to infinity. In this setting we also know that the gradient vanishes only at the unique minimiser w_* . This problem shows that gradient descent ‘works’ in the sense that $\min_{k=0,\dots,K} \|\nabla F(w_k)\|^2$ tends to zero when the number of iterations K tends to infinity. This problem also shows how the previous result can be obtained from this one.
- Problem 2: This problem adapts the results of Problem 1 to the setting of stochastic gradient descent which introduces randomness to improve the efficiency of the algorithm. Various behaviours are observed depending on the choice of step size (which can decay with the number of iterations).
- Problem 3: Deriving a Representer Theorem for every gradient descent iteration of least squares empirical risk minimisation, and explaining why this corresponds to a Representer Theorem. This shows that the weights obtained by gradient descent at each iteration are all linear combinations of the data points.

Problem 1

Suppose that a function $F: \mathbb{R}^N \rightarrow \mathbb{R}$ is M -smooth in the sense that

$$\|\nabla F(w) - \nabla F(z)\| \leq M\|w - z\|$$

for all $w, z \in \mathbb{R}^N$. Recall (from the lecture slides) that the Quadratic Upper-Bound Theorem leads to

$$F(z) \leq F(w) - \frac{1}{2M} \|\nabla F(w)\|^2 \tag{1}$$

and that the gradient descent is the sequence $(w_k)_{k=1}^K$ starting at w_0 and with the iteration

$$w_{k+1} = w_k - \frac{1}{M} \nabla F(w_k).$$

1) Prove that

$$\min_{k=0,\dots,K-1} \|\nabla F(w_k)\|^2 \leq \frac{2M}{K} (F(w_0) - F(w_*)).$$

2) Provide a condition on the number of iterations K (depending on ϵ) to guarantee that $\|\nabla F(w_k)\| \leq \epsilon$ for at least one iteration $k \in \{0, \dots, K-1\}$.

3) In the lecture slides, we proved that

$$F(w_K) - F(w_*) \leq \frac{M}{2K} \|w_0 - w_*\|^2.$$

Derive this inequality again starting from the result proved in **1)** under the assumption that $\|\nabla F(w)\|$ is upper bounded by a constant C for all $w \in \mathbb{R}^N$, and under the assumption that F satisfies the Polyak-Lojasiewicz inequality

$$\|\nabla F(w)\|^2 \geq 2\mu \left(F(w) - F(w_*) \right)$$

for some $\mu > 0$ and for all $w \in \mathbb{R}^N$, where we recall $w_* = \operatorname{argmin}_w F(w)$. We also assume that $\|w_0 - w_*\| \geq 2C/\mu$.

You may use the Mean Value Theorem which loosely states that, given $w, z \in \mathbb{R}^N$ there exists some $y \in \mathbb{R}^N$ such that $F(w) - F(z) = (w - z)^\top \nabla F(y)$.

Solution 1

1) Applying Equation (1) to the setting of gradient descent, we have

$$\begin{aligned} F(w_{k+1}) &\leq F(w_k) - \frac{1}{2M} \|\nabla F(w_k)\|^2 \\ \|\nabla F(w_k)\|^2 &\leq 2M \left(F(w_k) - F(w_{k+1}) \right) \end{aligned}$$

giving the telescoping sum

$$\begin{aligned} \sum_{k=0}^{K-1} \|\nabla F(w_k)\|^2 &\leq 2M \sum_{k=0}^{K-1} \left(F(w_k) - F(w_{k+1}) \right) \\ &\leq 2M \left(F(w_0) - F(w_K) \right). \end{aligned}$$

We deduce that

$$\begin{aligned} \min_{k=0, \dots, K-1} \|\nabla F(w_k)\|^2 &\leq \frac{1}{K} \sum_{k=0}^{K-1} \|\nabla F(w_k)\|^2 \\ &\leq \frac{2M}{K} \left(F(w_0) - F(w_K) \right) \\ &\leq \frac{2M}{K} \left(F(w_0) - F(w_*) \right) \end{aligned}$$

as $F(w_*) \leq F(w_K)$ since F attains its minima at w_* .

2) Note that

$$\min_{k=0, \dots, K-1} \|\nabla F(w_k)\|^2 \leq \frac{2M}{K} \left(F(w_0) - F(w_*) \right) \leq \epsilon^2$$

provided that $K \geq 2M \left(F(w_0) - F(w_*) \right) / \epsilon^2$, which means that $\|\nabla F(w_k)\| \leq \epsilon$ for at least one iteration $k \in \{0, \dots, K-1\}$.

3) Recall from **1)** that

$$\min_{k=0, \dots, K-1} \|\nabla F(w_k)\|^2 \leq \frac{2M}{K} \left(F(w_0) - F(w_*) \right).$$

By the Mean Value Theorem there exists some $y \in \mathbb{R}^N$ such that

$$F(w_0) - F(w_*) = (w_0 - w_*)^\top \nabla F(y),$$

the Cauchy–Schwarz inequality then gives

$$0 \leq F(w_0) - F(w_*) = (w_0 - w_*)^\top \nabla F(y) \leq \|w_0 - w_*\| \|\nabla F(y)\| \leq C \|w_0 - w_*\|.$$

This gives

$$\min_{k=0,\dots,K-1} \|\nabla F(w_k)\|^2 \leq \frac{2MC}{K} \|w_0 - w_*\|.$$

Using the Polyak-Lojasiewicz inequality, we obtain

$$\min_{k=0,\dots,K-1} 2\mu \left(F(w_k) - F(w_*) \right) \leq \frac{2MC}{K} \|w_0 - w_*\|.$$

Recall that

$$F(w_{k+1}) \leq F(w_k) - \frac{1}{2M} \|\nabla F(w_k)\|^2 \leq F(w_k)$$

so $F(w_K) \leq F(w_k)$ for $k \in \{0, \dots, K-1\}$, which gives

$$2\mu \left(F(w_K) - F(w_*) \right) \leq \min_{k=0,\dots,K-1} 2\mu \left(F(w_k) - F(w_*) \right) \leq \frac{2MC}{K} \|w_0 - w_*\|,$$

that is

$$F(w_K) - F(w_*) \leq \frac{MC}{K\mu} \|w_0 - w_*\|.$$

Finally, using $\|w_0 - w_*\| \geq 2C/\mu$ we get

$$F(w_K) - F(w_*) \leq \frac{M}{2K} \|w_0 - w_*\|^2.$$

Problem 2

In the classical setting of supervised learning, we have access to data points $((x_i, y_i))_{i=1}^n$, and we want to fit a predictor h_w (for example $h_w(x) = w^\top x$) which minimises the empirical risk

$$\frac{1}{n} \sum_{i=1}^n L(h_w(x_i), y_i) = F(w) = \frac{1}{n} \sum_{i=1}^n F_i(w).$$

Performing gradient descent gives $(w_k)_{k=0}^K$ where

$$w_{k+1} = w_k - \eta_k \nabla F(w_k) = w_k - \frac{\eta_k}{n} \sum_{i=1}^n \nabla F_i(w),$$

with positive η_k values. so for each iteration, n gradients need to be computed, which is expensive. To reduce the computational burden, for each iteration k , we can instead uniformly sample an index $r_k \in \{1, \dots, n\}$ and compute

$$w_{k+1} = w_k - \eta_k \nabla F_{r_k}(w).$$

This strategy is called stochastic gradient descend.

1) Prove that the stochastic gradient is an unbiased estimate of gradient.

2) Using the Quadratic Upper-Bound Theorem, and assuming $\mathbb{E} [\|\nabla F_{r_k}(w)\|^2] \leq \sigma^2$ for all w , show that

$$\mathbb{E} [F(w_{k+1}) \mid w_k] \leq F(w_k) - \eta_k \|\nabla F(w_k)\|^2 + \eta_k^2 \frac{M\sigma^2}{2}$$

where the expectation is with respect to r_k .

3) Prove that

$$\min_{k=0,\dots,K-1} \mathbb{E} [\|\nabla F(w_k)\|^2] \leq \frac{F(w_0) - F(w_*)}{\sum_{k=0}^{K-1} \eta_k} + \frac{M\sigma^2}{2} \frac{\sum_{k=0}^{K-1} \eta_k^2}{\sum_{k=0}^{K-1} \eta_k}.$$

4) Analyze the behaviour of the smallest value $\mathbb{E} [\|\nabla F(w_k)\|^2]$ for $k \in 0, \dots, K-1$ with respect to the number of iterations K using the $\mathcal{O}$ notation in the three different settings:

i) Constant step size $\eta_k = \eta$,

ii) Fastly decreasing step size $\eta_k = \eta/(k+1)$,

iii) Slowly decreasing step size $\eta_k = \eta/\sqrt{k+1}$.

You may use the facts that¹

$$\sum_{k=1}^K k^{-2} = \mathcal{O}(1), \quad \sum_{k=1}^K k^{-1} = \Theta(\log(K)), \quad \sum_{k=1}^K k^{-1/2} = \Theta(\sqrt{K}).$$

Solution 2

1) We have to show that $\mathbb{E} [\nabla F_r(w)] = \nabla F(w)$ where the expectation is over the (uniform) randomness of r in $\{1, \dots, n\}$. We have

$$\mathbb{E} [\nabla F_r(w)] = \sum_{i=1}^n \mathbb{P}(r = i) \nabla F_i(w) = \frac{1}{n} \sum_{i=1}^n \nabla F_i(w) = \nabla F(w).$$

2) The Quadratic Upper-Bound Theorem states that

$$F(z) \leq F(w) + \nabla F(w)^\top (z - w) + \frac{M}{2} \|z - w\|^2.$$

which gives

$$F(w_{k+1}) \leq F(w_k) + \nabla F(w_k)^\top (w_{k+1} - w_k) + \frac{M}{2} \|w_{k+1} - w_k\|^2.$$

Since in the stochastic gradient descent setting we have $w_{k+1} - w_k = -\eta_k \nabla F_{r_k}(w_k)$, we obtain

$$F(w_{k+1}) \leq F(w_k) - \eta_k \nabla F(w_k)^\top \nabla F_{r_k}(w_k) + \eta_k^2 \frac{M}{2} \|\nabla F_{r_k}(w_k)\|^2.$$

Taking the expectation conditioned on w_k (so the randomness comes from r_k which is uniformly sampled from $\{1, \dots, n\}$), we obtain

$$\begin{aligned} \mathbb{E} [F(w_{k+1}) \mid w_k] &\leq \mathbb{E} \left[F(w_k) - \eta_k \nabla F(w_k)^\top \nabla F_{r_k}(w_k) + \eta_k^2 \frac{M}{2} \|\nabla F_{r_k}(w_k)\|^2 \mid w_k \right] \\ &\leq F(w_k) - \eta_k \nabla F(w_k)^\top \mathbb{E} [\nabla F_{r_k}(w_k) \mid w_k] + \eta_k^2 \frac{M}{2} \mathbb{E} [\|\nabla F_{r_k}(w_k)\|^2 \mid w_k] \\ &\leq F(w_k) - \eta_k \nabla F(w_k)^\top \nabla F(w_k) + \eta_k^2 \frac{M}{2} \mathbb{E} [\|\nabla F_{r_k}(w_k)\|^2 \mid w_k] \\ &\leq F(w_k) - \eta_k \|\nabla F(w_k)\|^2 + \eta_k^2 \frac{M}{2} \mathbb{E} [\|\nabla F_{r_k}(w_k)\|^2 \mid w_k] \\ &\leq F(w_k) - \eta_k \|\nabla F(w_k)\|^2 + \eta_k^2 \frac{M\sigma^2}{2} \end{aligned}$$

¹See here and here.

using $\mathbb{E}[F_{r_k}(w_k) \mid w_k] = \nabla F(w_k)$ and $\mathbb{E}[\|\nabla F_{r_k}(w_k)\|^2 \mid w_k] \leq \sigma^2$.

3) Starting from

$$\mathbb{E}[F(w_{k+1}) \mid w_k] \leq F(w_k) - \eta_k \|\nabla F(w_k)\|^2 + \eta_k^2 \frac{M\sigma^2}{2},$$

we obtain

$$\eta_k \|\nabla F(w_k)\|^2 \leq F(w_k) - \mathbb{E}[F(w_{k+1}) \mid w_k] + \eta_k^2 \frac{M\sigma^2}{2}.$$

Taking the expectation of both sides, and using the law of total expectation $\mathbb{E}[\mathbb{E}[F(w_{k+1}) \mid w_k]] = \mathbb{E}[F(w_{k+1})]$, we obtain

$$\eta_k \mathbb{E}[\|\nabla F(w_k)\|^2] \leq \mathbb{E}[F(w_k)] - \mathbb{E}[F(w_{k+1})] + \eta_k^2 \frac{M\sigma^2}{2}.$$

As there is no conditioning, we stress that the expectation $\mathbb{E}[F(w_{k+1})]$ is taken with respect to all randomness (here incurring from the random draws of $r_0, \dots, r_k$ which determine w_{k+1}), and similarly for the other terms.

Summing over $k = 0, \dots, K-1$, and noting that the sum is telescoping, we get

$$\begin{aligned} \sum_{k=0}^{K-1} \eta_k \mathbb{E}[\|\nabla F(w_k)\|^2] &\leq \sum_{k=0}^{K-1} (\mathbb{E}[F(w_k)] - \mathbb{E}[F(w_{k+1})]) + \sum_{k=0}^{K-1} \eta_k^2 \frac{M\sigma^2}{2} \\ &= \mathbb{E}[F(w_0)] - \mathbb{E}[F(w_K)] + \frac{M\sigma^2}{2} \sum_{k=0}^{K-1} \eta_k^2 \\ &\leq F(w_0) - F(w_*) + \frac{M\sigma^2}{2} \sum_{k=0}^{K-1} \eta_k^2 \end{aligned}$$

as $\mathbb{E}[F(w_K)] \geq F(w_*)$ and $\mathbb{E}[F(w_0)] = F(w_0)$ as w_0 is deterministic. We now have

$$\begin{aligned} \left(\min_{k=0, \dots, K-1} \mathbb{E}[\|\nabla F(w_k)\|^2] \right) \sum_{k=0}^{K-1} \eta_k &\leq \sum_{k=0}^{K-1} \eta_k \mathbb{E}[\|\nabla F(w_k)\|^2] \\ &\leq F(w_0) - F(w_*) + \frac{M\sigma^2}{2} \sum_{k=0}^{K-1} \eta_k^2. \end{aligned}$$

Hence, we have

$$\min_{k=0, \dots, K-1} \mathbb{E}[\|\nabla F(w_k)\|^2] \leq \frac{F(w_0) - F(w_*)}{\sum_{k=0}^{K-1} \eta_k} + \frac{M\sigma^2}{2} \frac{\sum_{k=0}^{K-1} \eta_k^2}{\sum_{k=0}^{K-1} \eta_k}.$$

4) Recall that

$$\min_{k=0, \dots, K-1} \mathbb{E}[\|\nabla F(w_k)\|^2] \leq \frac{F(w_0) - F(w_*)}{\sum_{k=0}^{K-1} \eta_k} + \frac{M\sigma^2}{2} \frac{\sum_{k=0}^{K-1} \eta_k^2}{\sum_{k=0}^{K-1} \eta_k}.$$

i) Consider the case of constant step size $\eta_k = \eta$, so that

$$\begin{aligned} \min_{k=0, \dots, K-1} \mathbb{E}[\|\nabla F(w_k)\|^2] &\leq \frac{F(w_0) - F(w_*)}{\sum_{k=0}^{K-1} \eta_k} + \frac{M\sigma^2}{2} \frac{\sum_{k=0}^{K-1} \eta_k^2}{\sum_{k=0}^{K-1} \eta_k} \\ &= \frac{F(w_0) - F(w_*)}{K\eta} + \frac{M\sigma^2}{2} \frac{K\eta^2}{K\eta} \\ &= \frac{F(w_0) - F(w_*)}{K\eta} + \eta \frac{M\sigma^2}{2} \\ &= \mathcal{O}(K^{-1} + 1) \end{aligned}$$

which does not tend to 0 as K goes to infinity, that is, the stochastic gradient descend ‘does not work’ with constant step size: even in the limit $K \rightarrow \infty$ the minimum is not approached.

ii) Assume we have fastly decreasing step size $\eta_k = \eta/(k+1)$, then

$$\begin{aligned} \min_{k=0, \dots, K-1} \mathbb{E} [\|\nabla F(w_k)\|^2] &\leq \frac{F(w_0) - F(w_*)}{\eta \sum_{k=0}^{K-1} 1/(k+1)} + \frac{M\sigma^2 \eta^2 \sum_{k=0}^{K-1} 1/(k+1)^2}{2 \eta \sum_{k=0}^{K-1} 1/(k+1)} \\ &\leq \frac{F(w_0) - F(w_*)}{\eta \sum_{k=1}^K 1/k} + \eta \frac{M\sigma^2 \sum_{k=1}^K 1/k^2}{2 \sum_{k=1}^K 1/k} \\ &= \mathcal{O} \left(\frac{1}{\log(K)} + \frac{1}{\log(K)} \right) \\ &= \mathcal{O} \left(\frac{1}{\log(K)} \right) \end{aligned}$$

which converges to 0 as $K \rightarrow \infty$ but extremely slowly. We have used that $\sum_{k=1}^K k^{-1} = \Omega(\log(K))$ giving $\left(\sum_{k=1}^K k^{-1}\right)^{-1} = \mathcal{O}(1/\log(K))$, and $\sum_{k=1}^K k^{-2} \leq \sum_{k=1}^{\infty} k^{-2} = \frac{\pi^2}{6} = \mathcal{O}(1)$.

iii) Suppose we have slowly decreasing step size $\eta_k = \eta/\sqrt{k+1}$, then

$$\begin{aligned} \min_{k=0, \dots, K-1} \mathbb{E} [\|\nabla F(w_k)\|^2] &\leq \frac{F(w_0) - F(w_*)}{\eta \sum_{k=0}^{K-1} 1/\sqrt{k+1}} + \frac{M\sigma^2 \eta^2 \sum_{k=0}^{K-1} 1/(k+1)}{2 \eta \sum_{k=0}^{K-1} 1/\sqrt{k+1}} \\ &\leq \frac{F(w_0) - F(w_*)}{\eta \sum_{k=1}^K 1/\sqrt{k}} + \eta \frac{M\sigma^2 \sum_{k=1}^K 1/k}{2 \sum_{k=1}^K 1/\sqrt{k}} \\ &= \mathcal{O} \left(\frac{1}{\sqrt{K}} + \frac{\log(K)}{\sqrt{K}} \right) \\ &= \mathcal{O} \left(\frac{\log(K)}{\sqrt{K}} \right) \end{aligned}$$

which converges to 0 as $K \rightarrow \infty$ at a much faster rate than in (ii). We used the facts that $\sum_{k=1}^K k^{-1/2} = \Omega(\sqrt{K})$ giving $\left(\sum_{k=1}^K k^{-1/2}\right)^{-1} = \mathcal{O}(K^{-1/2})$, and $\sum_{k=1}^K k^{-1} = \mathcal{O}(\log(K))$.

Problem 3

In this problem we derive a Representer Theorem for every gradient descent iteration of least squares empirical risk minimisation.

1) *In the least squares setting with no bias term, we have some data $X \in \mathbb{R}^{n \times d}$, a label vector $y \in \mathbb{R}^n$, and that given some $x \in \mathbb{R}^n$ its label prediction is given by $w^\top x$ for some weight vector $w \in \mathbb{R}^n$. Give the expression for the least squares loss (with no bias), and write it in vector/matrix form.*

2) *Prove that all the iterations $(w_k)_{k=0,1,\dots}$ obtained by gradient descent can be written as $w_k = X^\top \alpha_k$ for some $(\alpha_k)_{k=0,1,\dots}$ each lying in $\mathbb{R}^n$.*

3) *Explain why $w_k = X^\top \alpha_k$ essentially corresponds to a ‘Representer Theorem’. Which kernel is used?*

Solution 3

1) In the least squares setting, we have some data $X \in \mathbb{R}^{n \times d}$, a label vector $y \in \mathbb{R}^n$, and given some $x \in \mathbb{R}^n$ its label prediction is given by $w^\top x$ for some weight vector $w \in \mathbb{R}^n$. The loss is then

$$F(w) = \frac{1}{n} \sum_{i=1}^n (w^\top x_i - y_i)^2 = \frac{1}{n} \sum_{i=1}^n \left(\sum_{j=1}^n w_j x_{i,j} - y_i \right)^2 = \frac{1}{n} \|Xw - y\|^2.$$

2) We prove this result by induction. This is, we first show that the result holds for $k = 0$. We then assume it holds for some $k \in \mathbb{N}$ and prove it then holds for $k + 1$.

For the initial case $k = 0$, we need to show that $w_0 = X^\top \alpha_0$ for some $\alpha_0 \in \mathbb{R}^n$. We derive an explicit form for α_0 as follows

$$\begin{aligned} X^\top \alpha_0 &= w_0 \\ XX^\top \alpha_0 &= Xw_0 \\ \alpha_0 &= (XX^\top)^{-1} Xw_0 \end{aligned}$$

where $XX^\top$ is the positive semi-definite kernel matrix which is invertible. Hence, letting $\alpha_0 = (XX^\top)^{-1} Xw_0$, we indeed have $w_0 = X^\top \alpha_0$ which proves the base case $k = 0$.

We now assume the case k and show that the case $k + 1$ holds. Assume that $w_k = X^\top \alpha_k$ for some $k \in \mathbb{N}$, we will show how to construct α_{k+1} such that $w_{k+1} = X^\top \alpha_{k+1}$. Using the gradient descent formula, we have

$$\begin{aligned} w_{k+1} &= w_k - \eta_k \nabla F(w_k) \\ &= w_k - \frac{\eta_k}{n} \nabla \|Xw_k - y\|^2 \\ &= w_k - \frac{2\eta_k}{n} X^\top (Xw_k - y) \\ &= X^\top \alpha_k - \frac{2\eta_k}{n} X^\top (XX^\top \alpha_k - y) \\ &= X^\top \left(\alpha_k - \frac{2\eta_k}{n} (XX^\top \alpha_k - y) \right) \\ &= X^\top \left(\left(I - \frac{2\eta_k}{n} XX^\top \right) \alpha_k + \frac{2\eta_k}{n} y \right) \\ &= X^\top \alpha_{k+1} \end{aligned}$$

where we have set $\alpha_{k+1} = \left(I - \frac{2\eta_k}{n} XX^\top \right) \alpha_k + \frac{2\eta_k}{n} y$.

We have proved by induction that w_k can be written in the form $X^\top \alpha_k$ for all $k \in \mathbb{N}$.

3) For a finite-dimensional feature map $\phi: \mathbb{R}^d \rightarrow \mathbb{R}^D$ such that $\phi(x) = (\phi_1(x), \dots, \phi_D(x))$ for $x \in \mathbb{R}^d$, the associated kernel is $k(x, y) = \phi(x)^\top \phi(y)$ for $x, y \in \mathbb{R}^d$.

A ‘Representer Theorem’ then loosely states that a solution in $\mathbb{R}^D$ is a linear combination of the feature map evaluated at the data points, that is, a solution is of the form

$$\sum_{i=1}^n \alpha_i \phi(x_i) = \begin{pmatrix} | & | & & | \\ \phi(x_1) & \phi(x_2) & \dots & \phi(x_n) \\ | & | & & | \end{pmatrix} \alpha \in \mathbb{R}^D$$

for some $\alpha \in \mathbb{R}^n$, and where the matrix displayed is of shape $D \times n$.

In the setting of this problem, we simply use the identity feature map $\phi(x) = x$ (and $D = d$), the kernel becomes the usual dot product $x^\top y$. We have shown that, for some $\alpha_k \in \mathbb{R}^n$, w_k can be expressed as

$$X^\top \alpha_k = \begin{pmatrix} | & | & & | \\ x_1 & x_2 & \dots & x_n \\ | & | & & | \end{pmatrix} \alpha_k = \sum_{i=1}^n (\alpha_k)_i x_i \in \mathbb{R}^d$$

which corresponds to a ‘Representer Theorem’ with linear kernel (*i.e.*, dot product).

Note that the Representer Theorem holds more generally for any kernel, including the ones with infinite-dimensional feature maps, in which case the Representer Theorem cannot be written in matrix form as presented here.